CSE 400: Fundamentals of Probability in Computing
Project Milestone 4

Topic: Multi-cycle Cyclostationary based Spectrum Sensing Algorithm for OFDM Signals with Noise Uncertainty in Cognitive Radio Networks

Group No.: 11
Wireless Communication

1 Baseline System and Motivation for Randomization

Brief Restatement of the Deterministic Baseline

In our earlier Milestone 3, the multi-cycle cyclostationary detector was implemented completely based on deterministic system. The signal which is received followed a binary hypothesis model, which is:

$$H_0 : x[n] = w[n]$$
$$H_1 : x[n] = s[n] + w[n]$$

The detector observes N different time baseband samples that is $x[n]$ and computes a test statistic T_{mc}. This T_{mc} is defined as the ratio of two cyclic autocorrelation energy terms:

$$T_{mc} = \frac{\text{Numerator}}{\text{Denominator}} = \frac{\frac{1}{N_\tau}\left|\sum_{n=0}^{N_\tau-1}\frac{1}{\eta_n}x[n]\,x[n+\tau]^*\sum_{k=1}^{K}e^{-j2\pi\alpha_k n}\right|^2}{\frac{1}{N_{\bar{\tau}}}\left|\sum_{n=0}^{N_{\bar{\tau}}-1}x[n]\,x[n+\bar{\tau}]^*\,e^{-j2\pi\beta n}\right|^2} \tag{1}$$

where

- $\tau = N_{\text{FFT}}$, $\{\alpha_k\}_{k=1}^{K}$ = the OFDM cyclic frequencies
- β = an arbitrary reference frequency
- K = the number of tested cyclic frequencies

Now, if $T_{mc} \geq \lambda$, then the system will give output that a signal is present. Otherwise, the system will give output that only noise is present. Now, from the probability of false alarm, the threshold is derived which is:

$$\lambda = \frac{1}{P_f} - 1$$

This decision will be fixed for all the trials, that will make the system deterministic. However, a very large number of Monte Carlo simulations are needed to estimate detection

probability at low SNR. This happens because the events of detection is rare and use of uniform sampling will not capture these events efficiently. Hence, the estimator will have high variance and the cost for computation will become very high. Hence, we need a randomized algorithm to solve this problem.

2 Randomized Algorithm Design

Decision Component Involving Randomness

Each Monte Carlo trial independently draws noise samples from the true noise distribution in the deterministic baseline. These noise samples are, $w[n] \sim \mathcal{CN}(0, \sigma^2)$.

Now, the sampling distribution itself is randomized. Hence, the noise samples are drawn from a proposal distribution $q(\cdot)$ instead of drawing from the true distribution $p(\cdot)$. The decision rule that is threshold comparison is unchanged. The randomization are completely based on how the input samples are generated for simulation.

Probability Distribution and Sampling Mechanism

Let us take $p(\mathbf{w})$ such that it is the true joint distribution of the N noise samples. A convenient proposal is a complex Gaussian with inflated variance:

$$q(\mathbf{w}) = \mathcal{CN}\left(0, \sigma_q^2 \mathbf{I}\right), \quad \sigma_q^2 > \sigma^2 \tag{2}$$

The inflation factor $\rho = \sigma_q^2/\sigma^2 > 1$ is selected so that the proposal places more probability mass in the tails that is in the region where T_{mc} remains close to or above the threshold λ. Now, for each simulated trial i, the ratio of likelihood can be computed as:

$$\tilde{w}_i = \frac{p(\mathbf{w}_i)}{q(\mathbf{w}_i)} = \left(\frac{\sigma_q^2}{\sigma^2}\right)^N \exp\left(-\frac{\|\mathbf{w}_i\|^2}{2}\left(\frac{1}{\sigma^2} - \frac{1}{\sigma_q^2}\right)\right) \tag{3}$$

The IS estimate of a probability $P = \mathbb{E}_p[\mathbf{1}\{T_{mc} > \lambda\}]$ is then the weighted average:

$$\hat{P}_{\text{IS}} = \frac{\sum_{i=1}^{M} \tilde{w}_i \, \mathbf{1}\{T_{mc}^{(i)} > \lambda\}}{\sum_{i=1}^{M} \tilde{w}_i} \tag{4}$$

where M is the number of IS trials which is much smaller than the number of standard Monte Carlo trials that are needed for the same estimation accuracy.

Difference from the Deterministic Baseline

- **Sample generation:** In the deterministic baseline, noise samples are taken directly from the true distribution. Hence, $p = \mathcal{CN}(0, \sigma^2)$. But in the IS algorithm, instead of taking samples directly from the true distribution, they are selected from a proposal distribution, which is, $q = \mathcal{CN}(0, \sigma_q^2)$ where $\sigma_q^2 > \sigma^2$, placing more mass in the tails of the distribution.

- **Trial weighting:** The baseline assigns equal weight $1/M$ to each and every trial. However, the IS algorithm assigns each trial an individual importance weight $\tilde{w}_i = p(\mathbf{w}_i)/q(\mathbf{w}_i)$. This will help to correct the shifted sampling distribution.

- **Decision rule and threshold:** Both the strategies use the same type of deterministic rule that is consider H_1 if $T_{mc} \geq \lambda$ otherwise consider H_0. Using this rule it then analytically derived the threshold $\lambda = 1/P_f - 1$.

- **Estimator bias:** Both the estimators are unbiased in different way. The IS estimator achieves unbiasedness through the likelihood-ratio reweighting which is shown in Equation (4), and not by changing the decision rule.

- **Estimator variance:** The key change is that the baseline estimator has high variance at low SNR. It is because detection events are rare under uniform sampling. And the IS estimator concentrates samples near the threshold. This significantly reduces the variance for the same number of trials M.

3 Probabilistic Reasoning Behind the Algorithm

How Randomness Addresses Uncertainty in the System

The fundamental difficulty faced in estimating P_f and P_d by using standard Monte Carlo is that these probabilities can be very small especially at high SNR. Hence, if we draw samples uniformly from the noise distribution then this means that most simulated trials will not produce a detection event. As a result of this a very few trials actually contribute information to the estimate. As a result, the resulting estimator has high variance and hence converges slowly.

The algorithm addresses this by redirecting randomness toward the informative region of the sample space. This is done by inflating the noise variance in the proposal distribution, which means IS generates more samples near and above the threshold λ. Each such sample carries a corrective weight $\tilde{w}_i$ which will account for the fact that it was oversampled.

The probabilistic correctness of the estimator is maintained because:

$$\mathbb{E}_p[\mathbf{1}\{T_{mc} > \lambda\}] = \mathbb{E}_q\left[\frac{p(\mathbf{w})}{q(\mathbf{w})}\mathbf{1}\{T_{mc} > \lambda\}\right] \tag{5}$$

This identity is true for any proposal q that has support wherever p is nonzero. This also guarantees unbiased estimation regardless of the choice of σ_q^2.

Probabilistic Assumptions Supporting the Design

1. **i.i.d. (Independent and Identically Distributed) Gaussian noise.** The noise samples $w[n]$ are i.i.d. $\mathcal{CN}(0, \sigma^2)$. This is the same assumption used throughout the project.

2. **Bounded noise uncertainty.** The actual noise variance is in the interval $[\sigma^2/\epsilon,\ \sigma^2\epsilon]$ and the actual noise variance is uniformly distributed over this interval. Here, $\epsilon = 10^{\Delta\sigma^2/10}$, and we can use IS for each noise-variance scenario independently.

3. **F-distribution of T_{mc} under H_0.** The test statistic is following an F-distribution when H_0 is true, which is proven in the baseline analysis. The IS weight calculation is based on the likelihood of the noise samples that follows a Gaussian distribution. So the result of closed-form F-distribution for λ is not disturbed.

4. **Absolute continuity.** The proposal distribution q must assign positive probability to all event to which p assigns positive probability as well. Since p and q are zero-mean complex Gaussians and $\sigma_q^2 > \sigma^2$, the tails of q are heavier than those of p, satisfying this condition.

Why This Randomized Strategy Is Expected to Improve Performance?

The variance of a Monte Carlo estimator of P_f or P_d is given by $\mathrm{Var}(\hat{P}) \propto P(1-P)/M$. When P is very small, the standard deviation of the estimator relative to P is large ($\sim 1/\sqrt{MP}$). IS can be used to reduce the effective variance of the Monte Carlo estimator by focusing the samples in the relevant region of the distribution. For spectrum sensing simulations, IS can be used to reduce the number of Monte Carlo trials by one or two orders of magnitude at low SNRs for which P_d needs to be estimated, such as -16.5 dB.

4 Implementation within the Simulation Pipeline

Modification of the Code Architecture

Our Milestone 3 codebase was cleanly modularized consisting of five core files: `signal_generator.py`, `noise_generator.py`, `cyclostationary_features.py`, `detector.py`, and `metrics.py`. In Milestone 4, we have included one new module which is `importance_sampling.py`. This file contained the IS-specific logic, and the existing five modules were not updated at all. The experiment script `run_simulation.py` has been expanded in a manner that it calls both the old Monte Carlo routines and the new IS routines, and both of them are run to completion so as to compare them directly.

The major design decision here was to maintain the detector pipeline itself the same. The `MultiCycleDetector` class, the cyclic calculation of autocorrelation, the threshold derivation, all that remains just the same. Only the wrapper around it becomes different, i.e. how noise is created and how the results of the detection are pooled together into probability estimates.

The Way the Implementation Generates Randomness

The noise in the baseline pipeline was sampled as $p = \mathcal{CN}(0, \sigma^2)$ and the trials were counted no less so towards the last estimate. The IS module brings in another noise source. Noise samples are selected from the proposal distribution:

$$q = \mathcal{CN}(0, \sigma_q^2), \quad \sigma_q^2 = \rho \cdot \sigma^2$$

in which $\rho > 1$ is a variance inflation factor. We have in our configuration file, $\rho = 3.0$, which indicates that the proposal noise is three times more than the true noise. This is implemented in the function `generate_proposal_noise()` inside `importance_sampling.py`. It is similar to the initial `generate_complex_gaussian_noise()` and instead of σ^2, it takes σ_q^2.

After drawing the proposal noise $\mathbf{w}_q$, the received signal is obtained as $\mathbf{x} = \mathbf{s} + \mathbf{w}_q$ (under H_1) or $\mathbf{x} = \mathbf{w}_q$ (under H_0), and the detector computes T_{mc} and makes a decision

in the usual way. After which the log-importance weight is calculated as:

$$\log \tilde{w}_i = N \log\left(\frac{\sigma_q^2}{\sigma^2}\right) - \|\mathbf{w}_i\|^2 \left(\frac{1}{\sigma^2} - \frac{1}{\sigma_q^2}\right)$$

It is this weight that puts right the fact that the noise has been sampled from q and not from p. The entire computation is performed in log-space and the log-sum-exp trick is used to avoid numerical overflow or underflow, which is important in this case since such weights can be large orders of magnitude when N is large.

The IS module plays three primary functions. `estimate_pfa_is()` is used when there are M trials under H_0 with proposal noise and returns the IS estimate of P_f along with its 95% confidence interval and ESS. `sweep_pd_vs_snr_is()` does it for P_d across a range of SNR values. And `compute_ess()` computes the effective sample size out of the log-weights.

What Has Not Changed since the Baseline

The remainder of the pipeline is the same as that found in Milestone 3:

- OFDM signal generation using QPSK modulation, IFFT, and cyclic prefix insertion
- Cyclic autocorrelation feature extraction at $\tau = N_{\text{FFT}} = 64$
- Multi-cycle test statistic computation (T_{mc} as a ratio of numerator and denominator CAC energies using $K = 5$ cyclic frequencies)
- Threshold $\lambda = 1/P_f - 1 = 9.0$ (for target $P_f = 0.1$)
- Deterministic decision rule: when $T_{mc} \geq \lambda$ then decide H_1, decide H_0 otherwise

The parameters of the simulation are also the same: FFT size is 64, CP length is 16, 1024 samples per trial, 1000 trials, SNR range -20 dB to 0 dB, and 1 dB of noise uncertainty. The IS variance inflation factor $\rho = 3.0$ is the only new parameter of the configuration and it was included in `config.json` as an `importance_sampling` block.

5 Assessment Measures and Comparative Outcomes

Evaluation Metrics

We apply five measures that can be used to compare the IS algorithm and the MC baseline. Each metric measures another entity of estimator quality, and the two of them combine to provide us with a complete picture of whether the randomization is of any benefit.

1. False Alarm Probability (P_f): This is a measure of the incidence of the detector falsely declaring H_1 when there is noise only. The theoretical value is $P_f = 1/(\lambda+1) = 0.1$. P_f estimates provided by both the MC baseline and the algorithm of IS should be near this value. When the IS estimate is not close to 0.1, it would indicate that the reweighting is introducing bias, which would be a serious problem. This measure is suitable since in spectrum sensing the false alarm rate must be kept under control. It is impossible to have the detector crying wolf too frequently.

2. Probability of Detection (P_d): This is the rate at which the detector correctly detects the signal in the real presence of it. We scan P_d through SNR values of -20 dB to 0 dB. Both algorithms are expected to give similar P_d curves provided that the reweighting of IS is working correctly. This is the key performance indicator for any detector because

at the end of the day, the entire purpose of the system is to identify the primary user signal.

3. 95% Confidence Interval Width: In the case of MC baseline, the CI is calculated analytically out of the binomial model:

$$\mathrm{CI}_{\mathrm{MC}} = 2 \times 1.96\sqrt{\frac{P_d(1-P_d)}{M}}$$

In the case of the IS algorithm, the CI is obtained as a result of the empirical variation of the self-normalized estimator. The closer CI of the same amount of trials implies that the estimator is more efficient, i.e. it provides us a more accurate answer using the same quantity of computational work. This measure will have a direct effect on the issue of whether IS will meet its objective of variance reduction.

4. Effective Sample Size (ESS): The ESS informs us of the number of M IS trials that are actually making useful contributions to the estimate. It is defined as:

$$\mathrm{ESS} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2}$$

When ESS $\approx M$, the IS proposal is suitable to the problem and the estimator acts almost like regular MC. When ESS $\ll M$, the weights are degenerate, that is, only a few of the trials have very large weights and the rest have negligible weights, and the estimator becomes unreliable. This measure can be used to diagnose the appropriateness of the decision made about ρ for our problem.

5. Concurrence with Theory: The closed-form expression $P_f = 1/(\lambda + 1)$ can be used as a ground truth which we can test. The bar chart comparing theoretical, MC, and IS estimates of P_f has a direct justification as to IS maintaining the statistical correctness of the detector. When all the three bars are near to 0.1, then we are assured that the system is in operation properly.

Comparative Results

Milestone 4 yields three additional comparison plots as a result of the simulation.

Plot 6 (P_d vs SNR with CI bands): This overlay indicates the MC baseline (blue, solid) and IS algorithm (green, dashed) each with shaded 95% confidence intervals around each curve. The curves are expected to match closely in the entire SNR range, which has it that IS does not corrupt the detection probability estimates. The CI bands let us comparatively test whether IS can give tighter estimates at any given SNR region. The IS CI band at low SNR when P_d is small might be smaller than the MC band, indicating that IS is performing its task of reducing variance on rare detection events.

Plot 7 (CI Width Comparison): This plot compares the entire width of the 95% confidence intervals from the two methods across SNR directly. At SNR points where P_d is middle range (0.3 to 0.7), the MC baseline CI is already relatively small due to the binomial variance $P_d(1-P_d)$ being maximised at around 0.5 and the estimator is effective in that region anyway. The IS benefit appears more at extreme P_d values (very low or very high), in which standard MC is not effective due to detection events being too rare or too common to make a precise estimate.

Plot 8 (P_f Bar Chart): The three bars — Theory, MC Baseline, and IS Algorithm — should all be close to 0.1. The plot is basically a sanity check. If the IS bar deviates

significantly from the theoretical value, it would indicate an issue with either the proposal distribution or the weight calculation. The error bar on the IS estimate shows its uncertainty range.

The first level of honesty that can be made here is that in case $P_f = 0.1$ and 1000 trials, the false alarm is not really so uncommon an event. False alarms will be caused in approximately 100 out of 1000 trials in normal MC, which is sufficient for a fair estimate. The IS advantage would become very intense indeed when we were working at $P_f = 0.001$ or less, with normal MC having to run tens of thousands of trials merely to chance enough false alarm events.

6 Lessons and Remaining Challenges

What Insights Did We Gain?

The greatest lesson that can be learned from this milestone is that Importance Sampling conserves the statistical characteristics of the detector but varies the way we can efficiently estimate its performance. The agreement between the theoretical value and both the MC and IS estimates of P_f at 0.1 confirms that the likelihood-ratio reweighting given by Equation (4) in fact works using finite samples. This is not a thing we ought to presume, as the self-normalized estimator is only asymptotically unbiased and because with $M = 1000$ we have a finite and somehow limited sample size.

The other observation is regarding the inflation factor ρ. We took $\rho = 3.0$, that is, we multiplied by three the noise power in the proposal. This moves the noise distribution such that there is a higher number of samples that fall when T_{mc} is within or above the threshold $\lambda = 9.0$. But this comes at a cost. The importance weights become increasingly variable with the value of ρ, and the ESS decreases. There is a sweet spot at which the proposal is aggressive enough to explore the interesting region without being too aggressive that the weights degenerate. Finding this sweet spot is problem-specific and something that you need to experiment with. For our setup, $\rho = 3.0$ gives a reasonable balance.

The third lesson is associated with the regime in which IS does bring a real benefit. In our present simulation model of $P_f = 0.1$ and medium SNR levels, detection events are not extremely rare. The MC baseline of 1000 trials already produces reasonable estimates that have reasonable confidence measures. The benefit of IS would be much more dramatic assuming that we wanted very small false alarm rates (such as $P_f = 0.001$ or $P_f = 0.0001$) or attempting to describe the performance of the detector at very low SNR (below -15 dB), in which case detections become truly seldom and standard MC requires an impractically large number of experiments merely in order to have a stable approximation.

In Which Cases Is IS Doing Better or Worse?

Under the following circumstances, IS is likely to do better than the baseline:

- **Rare event estimation:** In case P_f or P_d is very small, MC requires an enormous number of trials to make enough observations of detection events. IS focuses samples in the tail of the distribution, which gives credible estimates that require fewer trials. Think of it this way: in case you have to estimate a probability of 10^{-4}, standard

MC requires a minimum of 10^5 to 10^6 trials, but IS can get a reasonable estimate with 10^3 or 10^4 trials because it purposely creates samples from the region that is important.

- **Low SNR regime:** At SNR values of -15 dB or less, the signal is deeply buried in noise and detections become very rare occurrences. IS helps by amplifying the noise in the proposal, which is counterintuitive but makes these rare detection events more frequent under q. This artificial inflation is corrected by the importance weights.

Under the following circumstances, IS can give poorer results compared to the baseline:

- **High SNR regime:** When the signal is powerful, detections occur on nearly all tests under normal MC. Tilting the sampling distribution is unnecessary since the events of interest are already common. IS only increases extra weight variability in this case, and you end up with a noisier estimate than plain MC would give you.

- **Poorly chosen ρ:** When ρ is chosen too large, the proposal distribution becomes very different from the actual distribution. The result of this is that a small number of trials end up with very huge weights and most of them contribute very little. The ESS decreases dramatically and estimator variance can actually grow as compared to naive MC. This is the weight degeneracy problem, and this is the chief practical pitfall of IS.

Remaining Challenges

There are some open questions that can be further investigated.

To begin with, the value of $\rho = 3.0$ was fixed for all SNR values, although in principle the optimal inflation factor depends on the SNR. A large ρ may be required at very low SNR to push enough samples above the threshold, while at higher SNR a smaller ρ or even $\rho = 1$ (no IS) would be better. An adaptive scheme that tunes ρ per SNR point could improve overall efficiency.

Second, our current proposal only inflates the noise variance uniformly. A more advanced proposal could shape the distribution differently, for example by shifting the mean of the noise instead of merely scaling its variance, in order to better target the decision boundary of the detector.

Third, the IS framework was applied only to the noise generation stage of the pipeline. It can also be thought of to randomize other components of the system, such as using randomized detection thresholds or randomly selecting subsets of cyclic frequencies per trial, which would be a quite different form of algorithmic randomization entirely. These directions are left for future work.

End of Submission